



Enhancing geographical origin classification of Vietnamese oranges via hybrid preprocessing and supervised learning of UV-Vis spectral data

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ABSTRACT

Due to the high economic value of specific botanical varieties, origin fraud in oranges is commonly encountered through intentional mislabeling or misrepresentation. These fraudulent behaviors cause intricacies in quality control, quality assurance, and product reliability. This study proposes a simple, rapid, and non-destructive ultraviolet-visible (UV-Vis) spectroscopy method, accompanied by supervised learning models, to authenticate the geographical origins of Vinh oranges, one of Vietnam's premium sweet orange cultivars, collected from three key cultivation regions in northern Vietnam (Hoa Binh, Quang Ninh, and Tuyen Quang). To boost classification performance, we applied two preprocessing pipelines: standard normal variate and Savitzky-Golay smoothing (SNV+SG) and in combination with the first derivative (SNV+SG+FD). These were implemented prior to classification using supervised chemometrics (PCA-LDA), machine learning methods (random forest, RF), and deep learning models (artificial neural networks, ANN). Spectroscopic results demonstrated that electronic transitions ($\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$) observed within the 250–1000 nm range of 111 orange samples revealed the presence of quercetin, hesperidin, and β -carotene. Using minimal preprocessing, the overall discrimination efficiency of supervised models was remarkable (>90%) in the broadband range, while RF and ANN achieved acceptable accuracy (>80%) in the narrowband (250–400 nm). However, PCA-LDA yielded strong classification performance irrespective of the spectrum region and preprocessing level. Notably, the recognition ability of RF and ANN outperformed that of PCA-LDA on full-band spectra when data were further derived to first order, particularly ANN, with 100% of samples correctly classified. These findings emphasize the feasibility of this robust method for practical applications in orange origin classification and quality monitoring.

1. Introduction

Sweet oranges (*Citrus sinensis* L. Osbeck), commonly referred to as “oranges,” are among the most widely consumed citrus fruits worldwide, along with apples, watermelons, and bananas [1, 2]. Oranges are well-liked for consumption as fresh fruit or in juice, owing to their vibrant orange color, balanced sweet-tart taste, and pleasant aroma. It is worth highlighting that orange juice has been proven to offer numerous nutritional efficacies and health-promoting implications. It supports a healthy immune system, lessens dermal oxidative damage, and helps prevent cardiovascular and ocular diseases [3, 4]. Moreover, it has been implicated in mitigating the jeopardy of colorectal cancer, coronary diseases, and chronic asthma [5]. In addition, extracts from fresh oranges are considered effective in alleviating constipation and contribute to antioxidative, anti-inflammatory, anti-tumor, anti-proliferative, anti-fungal, anti-diabetic, and antithrombotic activities [2, 4, 6]. The aforementioned nutritional merits of oranges are primarily ascribed to their bioactive compounds, such as vitamin C (ascorbic acid), carotenoids, cyanidin glucoside, flavonoids (e.g., hesperidin, naringin), carbohydrates, and a complex mixture of organic acids (e.g., citric, malic, fumaric, tartaric, hydrocinnamic, ferulic), as indicated in Fig. 1 [5, 7]. Particularly, phenolic substances in oranges are potent antioxidative species and serve as bioactive scavengers of free radicals to inhibit the degradation of the cell membrane structure [4]. Thus, the health benefits of oranges are largely contingent upon their quality, which is defined by physical attributes (fragrance, color, and juicy pulp) and natural chemical elements.

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Due to the vast economic profits and high demand in the fruit beverage industry, the surge of food fraud and adulteration, involving oranges, has become ubiquitous around the world [8, 9]. The common fraudulent practices in food commodities are mainly performed by adding a co-fruit, which is less high-priced and more readily available, to the authentic juice [10]. For instance, adulterants/counterfeiters for orange juice could possibly be grapefruit, tangerine, mandarin, or even lemon. Additionally, the color and taste of orange juice are controlled and adjusted by inserting extraneous sugars, organic acids, water, and other additives [11]. These adulterations are perpetrated by the illegal food trade and violate the consumers' legitimate rights. Although these beverage frauds do not result in severe human health risks for consumers, intentional food adulteration consequently causes a loss of consumer confidence, harms the authenticity of the brand's origin image, and disrupts the overall balance of the commodity market with inequitable competition [12]. All of these factors emphasize the indispensable need of geographical origin determination and accurate origin labeling to avoid confusion and ensure the value of the certified food authenticity.

Different advanced instrumental methods are used to determine citrus fruit compositions, as reported in the literature [13-15]. To detect adulteration, mislabeling, and fraud, they are generally classified into two particular categories: targeted and untargeted analytical analyses. In targeted or fingerprinting analysis, these techniques focus on detecting and quantifying some specific, known markers. The typical marker compounds in citrus fruits are phenolic compounds, organic acids, and carbohydrates [5]. Although they require reference standards to acquire a priori information about the predefined analytes, high accuracy and sensitivity for known targets are a plus. Providing rapid and efficient separation of compounds, chromatographic-based techniques such as high-performance liquid chromatography, HPLC [16]; gas chromatography, GC [17, 18]; thin-layer chromatography, TLC [19]; and capillary electrophoresis, CE [20, 21], are potentially compatible with targeting the chemical composition for fruit authentication capabilities. In addition, some other advanced electroanalytical techniques have also been recently implemented [10]. On the other hand, untargeted approaches aim to characterize and detect all measurable compounds, including fingerprints and unknown ones, without bias. These discovery-based methods are well-suited to identify adulteration, novel markers, or unexpected contaminants in the food authenticity and forensics industry. The most commonly used ones are spectroscopic techniques, including vibrational spectroscopies such as near-IR (NIR) [11, 22] and Raman [23], electronic spectroscopies like fluorescence emission [24] and ultraviolet-visible absorption (UV-Vis) [25, 26], atomic emission spectroscopy (AES) [27], and nuclear magnetic resonance (NMR) [4, 28, 29]. Nevertheless, the common feature in both targeted and untargeted analyses is the demand for combining with data-driven approaches such as chemometrics (principal component analysis, PCA; hierarchical clustering analysis, HCA; partial least squares, PLS) or machine learning algorithms (supervised decision tree, DT; support vector machine, SVM; K-nearest neighbor, K-NN; random forest, RF; soft independent modeling of class analogy, SIMCA; artificial neural networks, ANN; etc.) [4, 9, 10]. They provide classification or clustering models and proper multivariate recognition in discriminating not only the authenticity and origin determination of fruits but also the quality control and different varieties of fruits from a high-dimensional and complex dataset.

Notwithstanding that classical physicochemical analyses could bear a cursory judgment on fruit classification, separation techniques (HPLC and/or GC) coupled with elemental methods (mass spectroscopy, MS) [6, 30-33] are deemed accurate, sensitive, and robust. Nevertheless, the demerits of targeted methods are that they are not cost-effective, are time-consuming, and laborious, and require a complicated derivatization stage of the samples, which possibly causes damage and ruins the integrity of the sample [34]. To be more precise, a significant amount of environmentally unfriendly organic solvents and high-cost reagents are used in chromatographic methods, while MS often entails another characterization method to couple with. NMR does not require extensive preparation, but its resource-intensive routine maintenance is its limitation. In comparison, UV-Vis spectroscopy, one of the linear spectroscopic tools, is recognized as a non-destructive, rapid, reproducible, and untargeted technique without the need for elaborate pretreatment [35]. Hence, it is tailored to meet the demands as an acceptable alternative to characterize and authenticate food samples such as honey [36-39], green coffee beans [40], roasted coffee [41], teas [42], *Centella asiatica* [43], and olive oils [44]. To date, this work is the first to apply the UV-Vis method integrated with machine learning specifically for identifying the production area of orange samples.

Oranges are citrus fruits that grow best in a subtropical to tropical climate, so the ideal regions for cultivating them include Brazil, the United States, China, India, and Mediterranean countries (such as Spain, Italy, Greece, and Morocco) [5, 45]. Among the top producers of oranges in the world, Vietnam is a promising country that has steadily increased its orange production in recent years (1.58 million tons in 2024, according to the U.S. Department of Agriculture) thanks to favorable tropical monsoon climate conditions. The primary orange-growing regions are Northern provinces such as Ha Noi, Hoa Binh, Hung Yen, Ha Giang, and Yen Bai, as well as Central provinces like Nghe An, Ha Tinh, and others. Vietnamese oranges are locally named "Cam," and Vietnam boasts several distinct varieties of oranges with unique characteristics: Sanh, Vinh, Cao Phong, Xa Doai, Khe May, Canh, among others. Therefore, verifying labeling integrity, e.g., "Sanh orange products of Hoa Binh province" should urgently be addressed. To the best of our knowledge, there has not been any report attempting to use qualitative spectroscopic data (electronic transitions) of orange biomarkers to differentiate the geographical origins of Vietnamese oranges.

Given the foregoing, the present study aimed to assess the potential of UV-Vis spectroscopy coupled with machine learning models for the classification of one particular Vietnamese orange variety (Vinh) according to their originating regions (Hoa Binh, Quang Ninh, and Tuyen Quang). To achieve this objective, the fingerprint absorption bands of the orange's chemical compositions (polyphenolics, flavonoids, and carotenoids) were first acquired and analyzed in a broader spectrum from the UV-Vis to NIR region to determine the analytically significant and data-rich spectral scanning range. Subsequently, supervised analytical techniques, including principal component analysis followed by linear discriminant analysis (PCA-LDA), machine learning (RF), and deep learning (ANN) algorithms, were combined to construct and generate predictive training models that could accurately define the cultivation location of oranges. By employing data preprocessing with smoothing, normalization, and first-derivative transformation, this study offers comparative results with the expectation to enhance the authenticity performance of farm products in general and orange fruits in particular.

2. Materials and methods

2.1. Reagents and chemicals

All chemicals were purchased from the Merck commercial supplier (USA) without further purification: dichloromethane (CH_2Cl_2 , $\geq 99.5\%$ purity) and ethanol (EtOH , $\geq 95\%$ purity). Double-distilled water with a resistivity of $18.2 \text{ M}\Omega\cdot\text{cm}$ was derived from the Milli-Q ultrapure water system at the National Institute for Food Control, Hanoi, and was carried out in all water-required chemical procedures.

2.2. Samples and sample preparation

All orange samples used in this study were of the *Citrus sinensis* variety and were harvested and collected from various regions of Vietnam from November 2023 until November 2024. A total of 111 samples with identified geographical locations were sourced from three provinces: Hoa Binh ($n = 54$), Quang Ninh ($n = 24$), and Tuyen Quang ($n = 33$), as shown in Fig. 2. All acquired oranges fulfilled the stipulated criteria of physical

characteristics: intact, unbruised, pest-free fruit peel, with no abnormal odor. These oranges were fully mature and ripe with a uniform color. There were no strange traces on the fruit peel that could be observed with the naked eye.

After orange samples were brought to the laboratory, they were washed, peeled, and carefully squeezed using an orange juicer to produce freshly squeezed orange juice or extracts. A strainer or sieve was utilized to discard the oranges' pulp and seeds. Orange extracts were centrifuged at 6,000 revolutions per minute (rpm) for 5 minutes and then filtered through a 0.45 μm polytetrafluoroethylene (PTFE) membrane to obtain the orange supernatant. Each supernatant was obtained and prepared in triplicate. The supernatant was then preserved and stored at 2–8 $^{\circ}\text{C}$ until further analysis by molecular UV-Vis spectroscopy.

2.3. Equipment and instrumentation

For the UV-Vis spectra measurement, a 1:20 dilution of the supernatant was conducted with a solvent mixture of CH_2Cl_2 : EtOH at a ratio of 2:3 (v/v). Aliquots (5 mL) of the juice-solvent mixture were loaded and placed in a quartz cuvette with an optical path of 10 mm before measuring the absorbance of the sample. Double-distilled water was used as a blank sample. All UV-Vis spectra acquisitions were performed at room temperature. The absorbance of the sample was measured using a Shimadzu U-3900 UV-Vis molecular absorption spectrophotometer (Hitachi, Japan) with a precision of ± 0.10 nm, connected to a computer loaded with PC OS Windows™ software. Each sample was scanned at a wavelength range of 250 nm to 1.00 μm , at 5-nm intervals. The recorded absorption spectra were subsequently exported to Microsoft Excel for statistical manipulation. Python version 3.11.0 version (The MathWorks, USA) was adopted to construct and perform the chemometric and artificial intelligence (AI)-driven models. To this end, a flowchart of the proposed data analysis procedure is depicted in Fig. 3.

2.4. Data processing

2.4.1. Data preprocessing

During the collection of UV-Vis spectra experiments, random noise and systematic errors, such as background interference and light scattering, are highly likely to contribute to the spectral information of orange samples. To enhance spectral resolution and optimize the prediction accuracy of supervised learning models from here on, we applied several preprocessing methods, including standard normal variate (SNV) normalization, Savitzky-Golay (SG) filtering, and the first-order derivative (FD) to eliminate the systematic variations in the baseline. One notable point is that SG was conducted by local polynomial regression to smooth high-frequency, inherently noisy data without distorting the peak shapes of spectral profiles [46]. Nonetheless, the physical differences between input spectral data (e.g., scattering effects and scale variation) were normalized by SNV to effectively ensure data comparability prior to modeling (PCA-LDA, RF, and ANN) [46, 47]. In principle, the SNV algorithm was standardized by mean subtraction and standard deviation division to achieve a mean of zero and a standard deviation of one. Furthermore, applying the FD to spectra aids in correcting baseline shifts, highlighting subtle peak detection, and honing classification or regression models [47]. Stemming from the best performances of the SNV and SG transformations [48], they were preferentially adopted in combination (SNV + SG) for all spectra processing. Another strategy involving FD pretreatment after SNV + SG (SNV + SG + FD) was established to compare the influences of more extensive pretreatment on models' accuracy against those with less pretreatment.

2.4.2. Principal component analysis (PCA)

Spectral datasets are often highly correlated, but the high-dimensional nature of the dataset poses many limitations for machine learning models. Among exploratory analyses for spectroscopy applications, multivariate analyses, e.g., PCA [49], two-dimensional correlation spectroscopy (2D-COS) [50], evolving factor analysis (EFA) [51], simple-to-use self-modeling mixture analysis (SIMPLISMA) [52], and multivariate curve resolution by alternating least squares (MCR-ALS) [53], are generally harnessed to define the number of significant features (band, image, chemical species, reactions, etc.). In particular, PCA is a preliminary unsupervised investigation tool to diminish data dimensionality, eliminate noise or outliers, and categorize sample clusters based on similarities and dissimilarities [49]. PCA is widely used to classify a complex dataset into reduced "latent" variables (PCs), which reflect the underlying structure of the original data. In the geometrical interpretation of PCA, data are first mean-centered and then projected onto the principal axis directions (loading vectors) of maximum variance (scores). The quantity of chosen PCs, thus, impinges upon the cumulative explained variance or scree plot (%) of the first few PCs. To avoid overfitting or losing crucial information, the number of selected PCs should explain at least 70% of the total variance. Otherwise, the principle of PCA was discussed in our previous work [54]. Therefore, the original dataset, a 111-sample $\times$ 161-wavelength matrix, in this study was either left unprocessed or subjected to the pretreatment using SNV + SG or SNV + SG + FD for all individual samples, and the average spectra of three replicates. Accordingly, this dataset was then passed through exploratory analysis using PCA procedures to analyze the key data features before applying supervised recognition models.

2.4.3. Supervised learning algorithms and model efficiency estimation

For model construction, the raw data were first mean-centered. After the data were preprocessed with SNV + SG or SNV + SG + FD, we proceeded to build supervised learning models for classification. The assayed supervised models were PCA-LDA, RF, and ANN. As a concept, RF is an ensemble learning algorithm that was introduced to boost predictive power by aggregating multiple decision trees [55]. On the other hand, ANN is a neuron system-simulated model constructed by interconnected layers (input, hidden, and output). Weights, activation functions, and biases are key factors used to model and capture nonlinear relationships in ANN [43]. In comparison, LDA is a linear algorithm characterized by an interpretable classification, whilst RF and ANN are nonlinear algorithms handling noise robustness and more complex patterns with higher accuracy but are more time-intensive. In the pursuit of dimensionality reduction and overfitting eradication, LDA was resolved by coupling with an unsupervised PCA technique. Compared to machine learning techniques like RF and ANN, PCA-LDA is a simpler, non-targeted multivariate analytical method.

To assess the effectiveness of the methodology, confusion matrices were generated, and model validation was determined via performance parameters: accuracy (A) or correct classification rate (CCR %), error rate (E), precision (P), recall or sensitivity (R), specificity (S), and $F1$ score ($F1$) [56, 57]. From a general perspective, a well-performing and efficient classification model is delineated by a high A , P , R , S , and $F1$, as well as a low E , and vice versa [58]. Those evaluation criteria (A , E , P , R , S , and $F1$) are calculated and presented in Equations 1, 2, 3, 4, 5, and 6, correspondingly.

$$A = \frac{TP + TN}{\text{Total}} \times 100 = \frac{TP + TN}{TP + FP + FN + TN} \times 100 (\%) \quad (1)$$

$$E = \frac{FP + FN}{Total} \times 100 = \frac{FP + FN}{TP + FP + FN + TN} \times 100 = 100 - A (\%) \quad (2)$$

$$P = \frac{TP}{Predicted\ positive} \times 100 = \frac{TP}{TP + FP} \times 100 (\%) \quad (3)$$

$$R = \frac{TP}{Actual\ positive} \times 100 = \frac{TP}{TP + FN} \times 100 (\%) \quad (4)$$

$$S = \frac{TN}{Actual\ negative} \times 100 = \frac{TN}{TN + FP} \times 100 (\%) \quad (5)$$

$$F1 = 2 \times \frac{P \times R}{P + R} \times 100 (\%) \quad (6)$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives from the confusion matrix in that order. In more detail, *A* quantifies the ratio of the total number of samples correctly predicted by the model (i.e., the sum of TP and TN) to the total number of samples (Equation 1). By contrast, *E* is calculated as the number of incorrectly classified samples (i.e., the sum of FP and FN) over the total number of samples (Equation 2). *P* measures the percentage of the number of samples correctly designated to the real class/population (TP) over the total samples appointed as that class (TP + FP), as shown in Equation 3. *R* is the TP rate, indicating the proportion of TP over the total quantity of samples in the actual positive class (TP + FN) (Equation 4). *S* or TN rate illustrates the fraction of the figure for samples correctly identified as not belonging to the corresponding class (TN) over the total number of samples in the actual negative class (TN + FP) (Equation 5). Lastly, *F1* (Equation 6) is depicted as a harmonic measure between *P* and *R* in assessing the efficiency of classification models when the dataset is imbalanced.

3. Results and discussion

3.1. UV-Vis spectral analysis of orange juice and spectra preprocessing

Spectral profiles of 111 orange samples from three Vietnamese provinces, spanning the UV to NIR range, are displayed in Fig. 4 (a: Hoa Binh, b: Quang Ninh, c: Tuyen Quang). The broader investigation domains, including UV-Vis (250-780 nm) and NIR (780-1000 nm), allow for the assessment of absorption peaks corresponding to an eclectic mixture of bioactive compounds (phenolics, flavonoids, vitamins, and carotenoids) present in orange fruit. It is evident from Fig. 4 that the key absorption bands are characterized at two specific wavelengths: 280 nm and 330 nm. The first band in the UV region can be ascribed to the electronic $\pi \rightarrow \pi^*$ orbital transitions of benzoyl-type aromatic ring systems and phenolic compounds (blue marks in Fig. 1) [59]. This peak is associated with the presence of flavonols (e.g., quercetin) and flavanones such as hesperidin and naringin [60]. On the other hand, vitamin C (ascorbic acid) has also been previously assigned to the 260-280 nm absorption band due to the enediol system with conjugated hydroxyls [61]. The absence of this peak in Fig. 4b, however, leads us to favor an interpretation attributing the signal to phenolic compounds.

Meanwhile, another characteristic UV absorption peak at 330 nm is traced to the localized transitions of cinnamoyl ring-related systems like quercetin (red mark in Fig. 1) [59, 62]. This aromatic system refers to a benzene ring linked to a three-carbon α , β -unsaturated carbonyl group ($-\text{CH}=\text{CH}-\text{CO}-$) with $\pi \rightarrow \pi^*$ transitions of the benzene ring and $n \rightarrow \pi^*$ transitions of carbonyl groups ($\text{C}=\text{O}$). Nevertheless, most UV-Vis spectra of orange samples from Quang Ninh Province (Fig. 4b) do not indicate the presence of the lower band, and this suggests that the peak at around 330 nm may be associated with cinnamoyl groups found in similar compounds such as conjugated lactones (coumarins) [61]. Broadly, typical orange juice contains a minimal quantity of natural coumarins (auraptene), primarily in the peel [63, 64]. In addition, there are some minor recognitions of absorption bands at 450 nm in Tuyen Quang's orange samples (Fig. 4c). This is attributed to the presence of carotenoids (e.g., β -carotene), naturally occurring pigments with long π -conjugated chains [61, 65]. These compounds absorb in the visible region by virtue of $\pi \rightarrow \pi^*$ transitions and are fundamentally responsible for the orange color. Gebishu *et al.* [61] and Zeb *et al.* [66] also reported dominant absorption peaks around 390 nm and 680 nm, resulting from particular signals of vitamin A and chlorophyll in fresh orange fruit juice samples, respectively, whereas we did not detect these characteristic bands in our samples.

Overall, the UV-Vis data of orange samples from three distinct provinces, especially Hoa Binh and Tuyen Quang, exhibited a similar spectral pattern but differed solely in the absorbance at each wavelength. This observation is tantamount to the equivalence of primary chemical compounds in oranges from different locations. To elaborate, the observed spectra provide complex insights into the chemical composition of oranges through multiple characteristic absorption bands of certain antioxidants. Whilst the pronounced UV absorption (< 400 nm) arises from $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions of phenolic compounds, flavonoids, vitamins, and coumarins, conjugated systems such as carotenoids (provitamin A precursors) absorb in the visible range around 400-500 nm on account of $\pi \rightarrow \pi^*$ transitions. A few published articles [67, 68] observed similar peaks in their orange-based spectra, but no specific designation of these UV absorption bands was highlighted. This study, thus, could also serve as a cross-reference to determine the absorption peaks of oranges for other ongoing studies. To verify and fortify our conclusion, other chemical characterization results via FT-IR spectroscopy, NMR, and quantitative analysis by HPLC will be presented in the next publication. In addition, the UV-Vis analyses appear to be largely unaffected by the spectral regime above 400 nm. Therefore, whether it is superfluous or not to utilize the data of complete wavelength coverage for detecting chemical components in oranges and determining their location of origin will be quantified via models' accuracy metrics in the next subsection 3.3.1.

In spite of evincing discrepancies in the acquired UV-Vis spectra among various orange populations from distinct places of origin, it is impossible to distinguish geographical provenance based on the raw UV-Vis spectra in Fig. 4. Hence, further data mining and data-driven assessment are propounded as primary complementary approaches to extract and elucidate the class differentiation. A data preprocessing step using SNV + SG + FD was endeavored to make all spectra comparable in terms of intensities and improve the reliability of the spectral analysis. The spectral correction employing preprocessing techniques (Fig. 5) revealed a fingerprinted peak shape of cinnamoyl entities at the band around 330 nm.

3.2. Exploratory analysis by PCA

To comprehend the structure and characteristics of the dataset in more detail, we utilized PCA multivariate analysis on the raw data and preprocessed data (SNV + SG and SNV + SG + FD), depicted in Fig. 6. Apart from the purpose of data visualization and dimensionality reduction, we aimed to achieve better cluster separation with preprocessed data compared to raw data. Firstly, the two-dimensional PCA score plot analysis of unsmoothed input data (Fig. 6a) yielded results showing that the first two PCs explained 85.4% of the total cumulative variance. Thus, the separation of the three groups of samples (Hoa Binh, Quang Ninh, and Tuyen Quang) could not be completed under the unsupervised scenario, and no single PC was sufficiently suitable to discriminate orange populations according to their origins. It can also be seen that sample clustering exists in the

three orange groups, but with poor visual clarity. The superposition and scattering occurred between samples of the three clusters, which may be a result of background noise, making the distinction of orange juice by region less accurate.

To improve the dispersion of the points and clustering trends, PCA was performed again with preprocessing strategies (Fig. 6b and 6c). The combination of PC1 and PC2 from SNV + SG and SNV + SG + FD pretreatment accounted for 93.5% and 88.4% of the total variance, respectively. Interestingly, the separation in PCA results with these preprocessing methods is still limited, and we could not completely differentiate the three origins of orange samples even with preprocessing techniques. However, it is worth mentioning that orange samples from different origins show a certain clustering pattern in the PCA score plots. In particular, as shown in Fig. 6b, Hoa Binh samples are principally located at the positive values of PC1, while samples from Quang Ninh are primarily distributed in the negative part of PC1. In comparison with Hoa Binh and Quang Ninh orange samples, Tuyen Quang samples exhibited a distinct scattering pattern across quadrants, suggesting a resemblance in chemical ingredients and orange content in this region with Hoa Binh and Quang Ninh samples. This observation could be derived from the geographical reason, which is that Tuyen Quang is a northern mountainous area situated between Hoa Binh and Quang Ninh Provinces (see Fig. 2). Therefore, a slight discrepancy in chemical composition (peculiarly the absorption bands at 450 nm) was possibly traced in samples from Tuyen Quang Province.

3.3. Discrimination of the geographical origin of oranges by supervised learning techniques

3.3.1. Effect of wavelength selection on the recognition ability of classification models

For the discrimination of geographical origins of oranges, the common dataset of 111 orange samples, after data pretreatment, was randomly divided into two data subsets: the training/modeling set for constructing and training the models (80% sample ratio, $n = 88$) and the prediction/test set with true positives for evaluating the performance of the trained models (20% sample ratio, $n = 23$). The label data was encoded into three equivalent types: Hoa Binh, Quang Ninh, and Tuyen Quang.

Despite efforts to reveal a significant separation of geographical origins through PCA with preprocessed data, none of the PCA results indicated a satisfactory enhancement in geographical classification. At this point, it is imperative to highlight the importance of supervised techniques, i.e., discriminant methods such as PCA-LDA, machine learning methods such as RF, and deep learning approaches like ANN, in discriminating the geographical origins of oranges. As a consequence of insufficient stability and significant error in classification analyses without data preprocessing steps, we solely assayed supervised techniques with pretreated data. Furthermore, we extended our analysis beyond the traditional UV-Vis range into the NIR region to compare how spectral regions affect model performance. To be more specific, we also evaluated the performance of these models in the narrowband range (250–400 nm) to validate the necessity to adopt the full range of wavelengths (250–1000 nm) for classification effectiveness. The overall performances of these classification algorithms are summarized and compared in Table 1 (A_{training} and A_{test} are the classification accuracies of the training and test sets, respectively. Other parameters are the results from the test set unless otherwise stated).

As indicated by the A -index results (Equation 1) comparing the UV spectral range (250–400 nm) with the entire spectral range (250–1000 nm) in Table 1, it can be seen that utilizing the full-band spectral data in the recognition application of oranges presents a more accurate analysis in predicting and labeling the oranges' derivation in this study. To exemplify, the aggregate recognition rate of the supervised classification techniques on the prediction set using the all-band spectral matrix did not fall below 90% with SNV + SG pretreatment and was just under 95% with the SNV + SG + FD procedure (see Table 1). The less accurate performance of classification results in the subset of spectral bands, represented in Figs. S1 and S2, was derived from results of machine learning techniques (RF and ANN). The rationale for this observation can be ascribed to the characteristics of the spectral data and the behavior of the models in recognition tasks. In general, RF and ANN are two complex nonlinear classifiers that require large datasets and fine-tuned hyperparameters (e.g., number of trees, depth, etc. for RF, and learning rate, hidden neuron layers, activation functions, etc. for ANN, see Fig. 3) [69]. Therefore, when executed in a smaller or moderately sized [m samples $\times$ n wavelengths] matrix, RF may suffer from overfitting, and the ANN model is more prone to hyperparameter misconfiguration.

3.3.2. Performance comparisons among supervised classification algorithms

From the prediction performance on the test set (Table 1), PCA-LDA achieved the highest classification accuracy with 100% in both pretreatments when the data was merely selected in the UV range (250–400 nm region). The number of PCs selected to construct the PCA-LDA model and ensure that the obtained classification error rate was minimized was 10. In comparison to PCA-LDA, the performance of both machine learning techniques, RF and ANN, was 13.1% lower when preprocessed with SNV normalization and SG smoothing. However, the accuracy of these two techniques ameliorated, in particular, to 91.3% for RF and 95.6% for ANN, when the dataset was further treated with FD. The superior performance of PCA-LDA over machine learning methods in this scenario is due to the fact that a coupling of PCA with LDA discriminates more efficiently on a smaller data size and favors the elimination of the overfitting phenomenon [70, 71].

Grounded in the satisfactory efficiency of predictive power (A) of supervised models on the full UV-Vis band, the other performance parameters (E , P , R , S , and $F1$) were determined in Table 1, and they were calculated from confusion matrices (Fig. 7). When adopting the dataset pretreated with SNV + SG, the discrimination results reveal that all three algorithms—PCA-LDA, RF, and ANN—offer sufficient classification accuracy, specificity, and $F1$ score ($> 90\%$). In the test set ($n = 23$), the PCA-LDA algorithm still provides better classification performance for oranges in three provinces, with an accuracy of 95.6%; the remaining two algorithms, RF and ANN, achieve lower accuracies of 91.3%. To specify, the PCA-LDA model incorrectly predicted one Hoa Binh orange sample as originating from Tuyen Quang Province; on the contrary, both RF and ANN mutually misclassified one sample from Hoa Binh or Quang Ninh Province as the other (Figs. 7a, 7b, and 7c). This observation evinces that Tuyen Quang, Hoa Binh, and Quang Ninh orange samples contain similar chemical components and thereby share common UV-Vis spectral features. Moreover, the justification for the prediction ability of supervised models here is still associated with the size of the dataset: the low-parameter PCA-LDA algorithm is simpler and more effective with high-dimensional matrices when the sample size is not large [71]. On the other hand, the complexity of the RF and ANN models, with more training samples, could overcomplicate the decision boundaries and easily underperform compared to a linear classifier such as PCA-LDA [72]. Meanwhile, the RF model classified all orange samples correctly in the training set, as described in Fig. S3, while ANN and PCA-LDA inaccurately predicted 1.20% and 3.50%, respectively. It is obvious that machine learning models outperformed PCA-LDA in the recognition rate on the training set, which exhibits a reversed tendency in comparison to the test set. The alteration in classification efficiency is considered to stem from the larger matrix of the dataset in the training set ($n = 88$), which is favorable to the working principle of more intricate models like RF or ANN.

Furthermore, FD was further applied after smoothing and normalization to validate the advantages of derivative preprocessing and examine its effect on origin traceability. Figs. 7d, 7e, and 7f illustrate the confusion matrix classification results obtained from PCA-LDA, RF, and ANN on the test set of full-band spectral data with the pretreatment of SNV + SG + FD, respectively. It is evident that the recognition ability of PCA-LDA remained intact in both test and training sets when data was pretreated further with the first-order derivative (Table 1). Conversely, the machine learning

model's predictive ability was enhanced significantly after the data was differentiated once, notably in the case of the ANN algorithm. In detail, RF succeeded in making a clear discrimination with up to 95.6% accuracy in the test set, and ANN acquired a complete separation of 11 Hoa Binh samples, 5 Quang Ninh samples, and 7 Tuyen Quang samples without the presence of false negative or false positive samples. In the training set, ANN also attained a zero error rate in distinguishing orange samples from these three provinces (see Fig. S4). The absolute 100% accuracy, specificity, and F1 score of the ANN model demonstrated that none of the Vietnamese orange samples were wrongly identified nor inaccurately classified. These results highlight the excellent and elevated distinction performance of ANN and RF machine learning models under the first-order derivative pretreatment of the spectral dataset over the simpler, linear PCA-LDA classifier, even though the dataset size remained identical to the case without FD pretreatment. This transformation could be deciphered by the technical interpretation of data pretreatment with FD. When the spectral data was enhanced with FD, the derivative-transformed data shifted toward a richer representation and captured nonlinear structures [72, 73]. Whilst PCA-LDA is predicated on optimal linear separability, this data derivative was responsible for the outperformance of complex nonlinear models like RF and ANN in terms of capturing nonlinear class boundaries. Between the two machine learning models, ANN slightly surpassed RF in nonlinearity handling owing to its better tuning and flexible learning of temporal dynamics or continuous patterns [74, 75]. Taken together, these results indicated that although the PCA-LDA algorithm sufficed to predict and recognize the geographical origins of oranges in a lower-dimensional dataset, data pretreatment with a combination of SNV, SG, and FD greatly enhanced the discrimination efficiency of the two machine learning models, RF and ANN.

In this context, it is essential to emphasize that empirical UV-Vis spectral data reflect sufficient chemical components to permit the development of supervised algorithms for the differentiation of Vietnamese oranges according to their originating locations. Even though machine learning methods correctly identified more than 80% of the orange samples in the selective partial-band spectral data, these accuracies generally met satisfactory classification thresholds and acceptance criteria for discrimination [76]. All models with *A*, *E*, *P*, *R*, *S*, and *F1* values above 90% in full-spectrum analysis were considered high-performing with respect to their recognition rate. To reinforce and optimize the models' efficiency, increasing the number of orange samples (*n*) from different orange cultivars, rather than just Vinh orange, in specific orange seasons should be further examined in the next study. In comparison with other investigations in the literature [10, 45, 48, 77, 78], their method exhibited good classification results for oranges' areas of origin. Nevertheless, utilizing UV-Vis spectroscopy coupled with supervised learning models underscores several merits related to its cost effectiveness, simplicity, and rapidity. Overall, the satisfactory accuracy, precision, sensitivity, and specificity verified the validation of this coupled method in the geographical provenance identification of Vietnamese oranges in both routine controls and large-scale industry standards.

4. Conclusions

To address the global challenge of food fraud and adulteration, this study validates the capability of UV-Vis spectroscopy as a rapid, cost-effective, and efficient analytical method for origin traceability of Vietnamese orange juice. Combined with advanced chemometric (PCA-LDA) and AI-driven models (RF and ANN), we successfully discriminated and assigned 111 orange samples to their province of provenance among three northern provinces: Hoa Binh, Quang Ninh, and Tuyen Quang. This untargeted spectroscopic method was able to detect several key electronic transitions ($\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$), associated with flavonoids and carotenoids in the oranges' spectral features. These findings revealed the potential of UV-Vis spectroscopy for capturing relevant chemical markers in oranges. Intriguingly, exploratory analysis by PCA alone displayed ambiguous cluster separation between samples and could not entirely separate or distinguish their cultivation region, even after the data were smoothed, normalized, and first-order derived. Consequently, these results underscore the inevitability of supervised learning adoption for high-performing classification. This study demonstrated that geographical origin authentication using PCA-LDA was completely achievable by utilizing only UV-Vis data, avoiding the need to extend into the NIR region, due to the striking absorption band mainly located in the UV region. Nonetheless, the two AI-based models achieved superior discrimination, exceeding 90% accuracy in the broadband spectrum, indicating that these methods yielded better prediction ability with large-scale and complex datasets, albeit within regions devoid of spectral characteristics. The predictive power of machine learning models exhibited a substantial enhancement when further implementing the first-order derivative with smoothed and normalized data. The ANN model, in particular, delivered an absolute distinction of all orange samples with a 100% correct classification rate. In summary, this study proposes UV-Vis spectroscopy, in conjunction with supervised learning models, as a powerful and practical classification approach for the geographical provenance of Vietnamese oranges. These findings support using this proposed method in industry applications for improved quality assurance, quality control, and product integrity in the food sector.

CRedit authorship contribution statement

Duc Thanh Nguyen: Investigation, Data curation, Methodology, Writing-review and editing. **Manh Son Nguyen:** Investigation, Data curation, Formal analysis, Software, Visualization, Writing-review and editing. **Huu Vang Pham:** Formal analysis, Software. **Manh Ha Nguyen:** Resources. **Duc Phong Nguyen:** Investigation, Data curation, Formal analysis, Software, Visualization, Writing-review and editing. **Thi Van Anh Nguyen:** Investigation, Formal analysis. **Thi Hong Hao Le:** Methodology, Validation. **Ba Lich Pham:** Writing-original draft, Writing-review and editing, Investigation, Visualization, Validation. **Thi Thao Ta:** Conceptualization, Methodology, Validation, Writing-review and editing, Funding acquisition, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data may be shared upon request and authorization from the corresponding authors.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:.....

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Tables

Table 1. Classification performance parameters (%) of PCA-LDA, RF, and ANN models

Wavelength range (nm)	Pretreatment	Performance evaluation (%)	Classification models		
			PCA-LDA	RF	ANN
250-400	SNV + SG (order = 0)	A_{test}	100	86.9	86.9
		$A_{training}$	98.8	100	95.4
	SNV + SG + FD (order = 1)	A_{test}	100	91.3	95.6
		$A_{training}$	98.8	100	97.7
250-1000	SNV + SG (order = 0)	A_{test}	95.6	91.3	91.3
		$A_{training}$	96.5	100	98.8
		E	4.40	8.70	8.70
		P	96.5	90.9	90.9
		R	97.7	90.9	90.9
		S	98.8	90.9	95.6
		$F1$	96.5	90.9	90.9
	SNV + SG + FD (order = 1)	A_{test}	95.6	95.6	100
		$A_{training}$	96.5	100	100
		E	4.40	4.40	0
		P	96.5	94.4	100
		R	97.7	97.7	100
		S	98.8	98.8	100
		$F1$	96.5	95.6	100

Figures

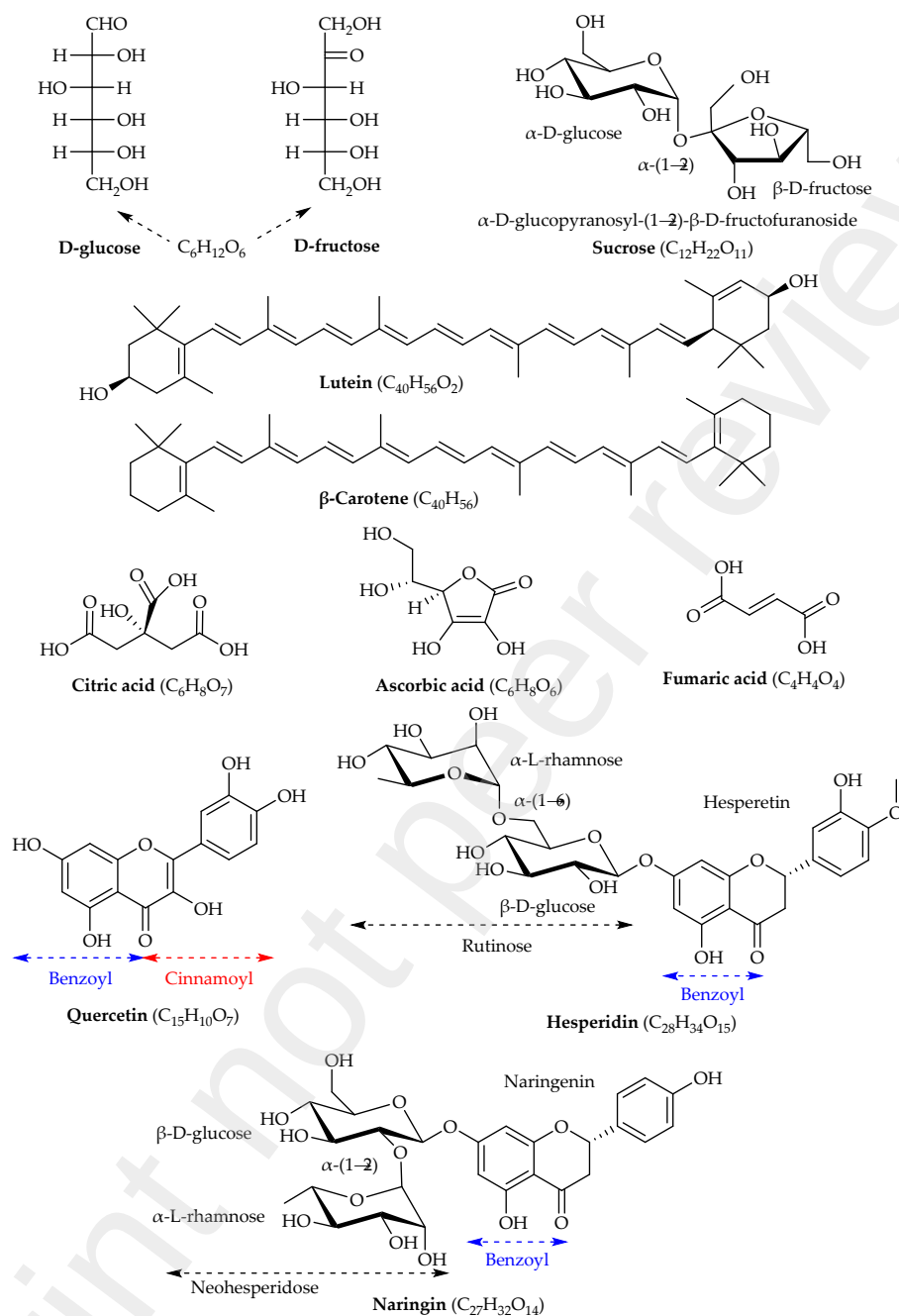


Fig. 1. Chemical structures of sugars (glucose, fructose, and sucrose/saccharose), carotenoids (lutein, β -carotene), organic acids (citric, fumaric, ascorbic acids), and flavonoids (quercetin, hesperidin, naringin) present in oranges.

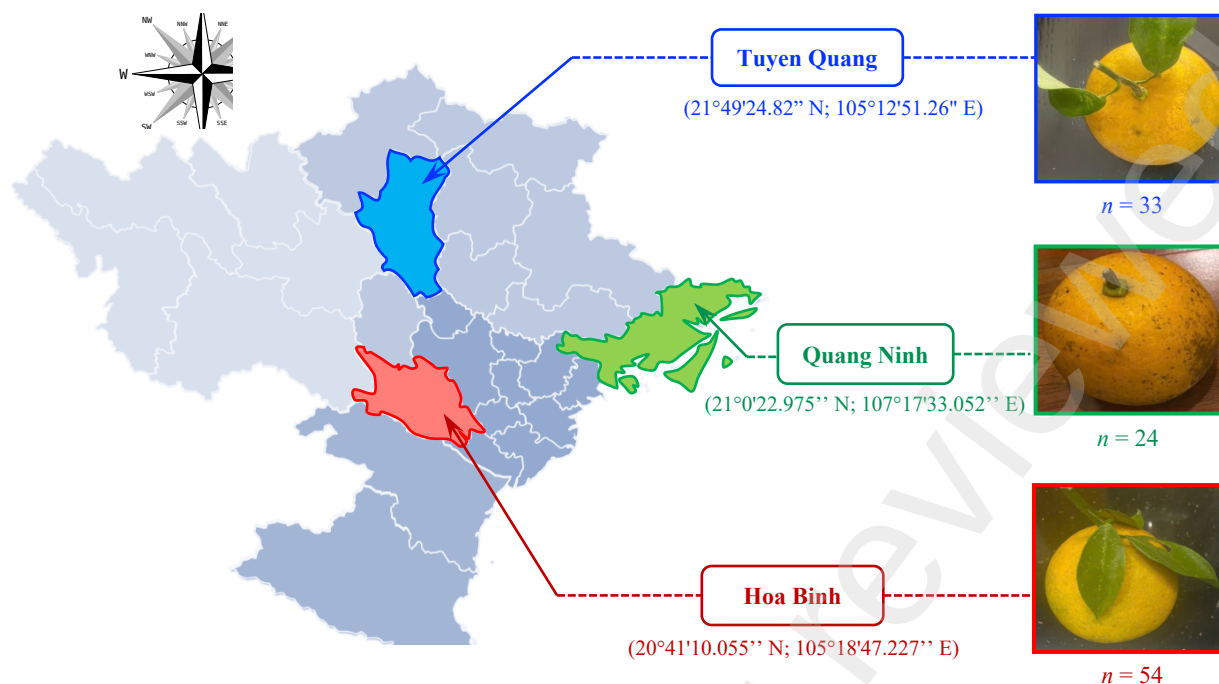


Fig. 2. Schematic diagram of the geographical distribution and representative orange samples from three different northern regions (Tuyen Quang, Quang Ninh, and Hoa Binh), along with their corresponding coordinates in Vietnam. The n value designates the number of collected orange samples from each northern province.

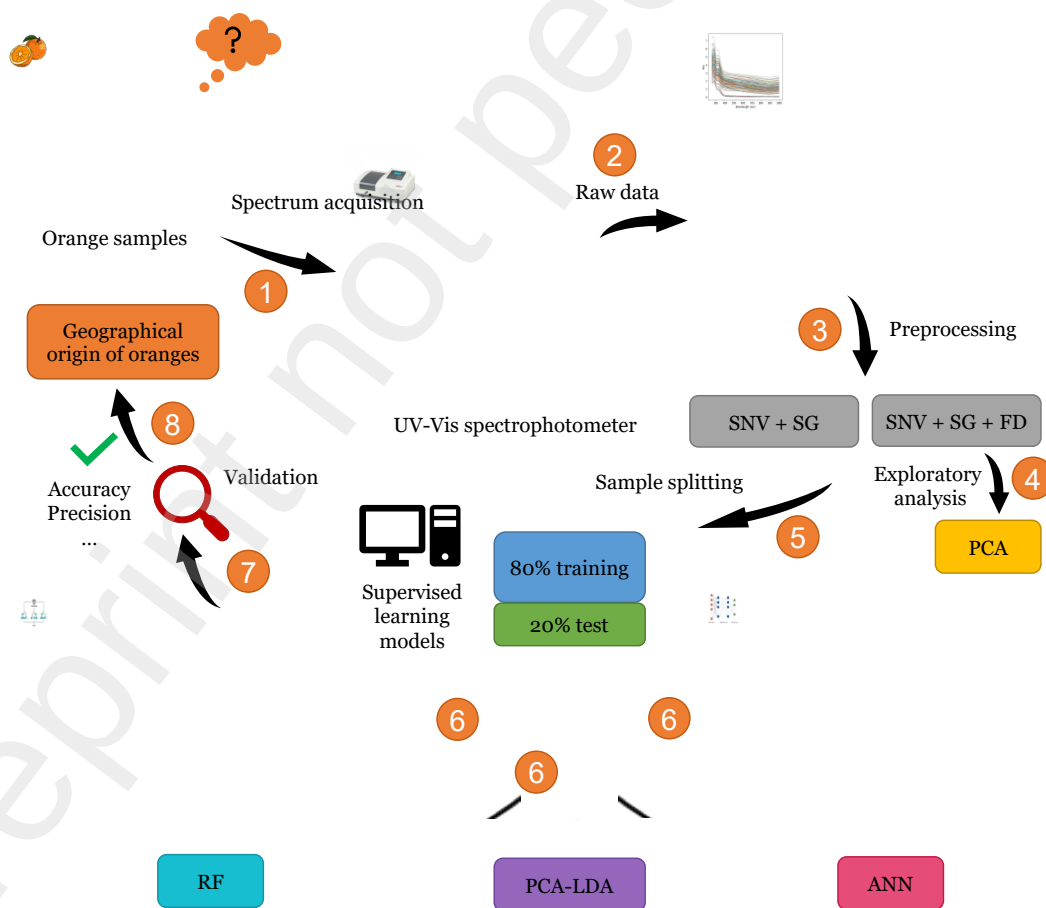


Fig. 3. Overview flowchart for supervised learning models of UV-Vis fingerprints of orange samples. (1) Spectra acquisition after orange samples were treated and prepared, (2) Spectra plotting based on raw data, (3) Preprocessing with raw data by two pipelines: SNV + SG or SNG + SG + FD, (4) Exploratory analysis by PCA, (5) Sample splitting for supervised learning models (RF, PCA-LDA, and ANN), (6) Classification performance of supervised learning models, (7) Model validation by different parameters, and (8) Identification of geographical origin of oranges.

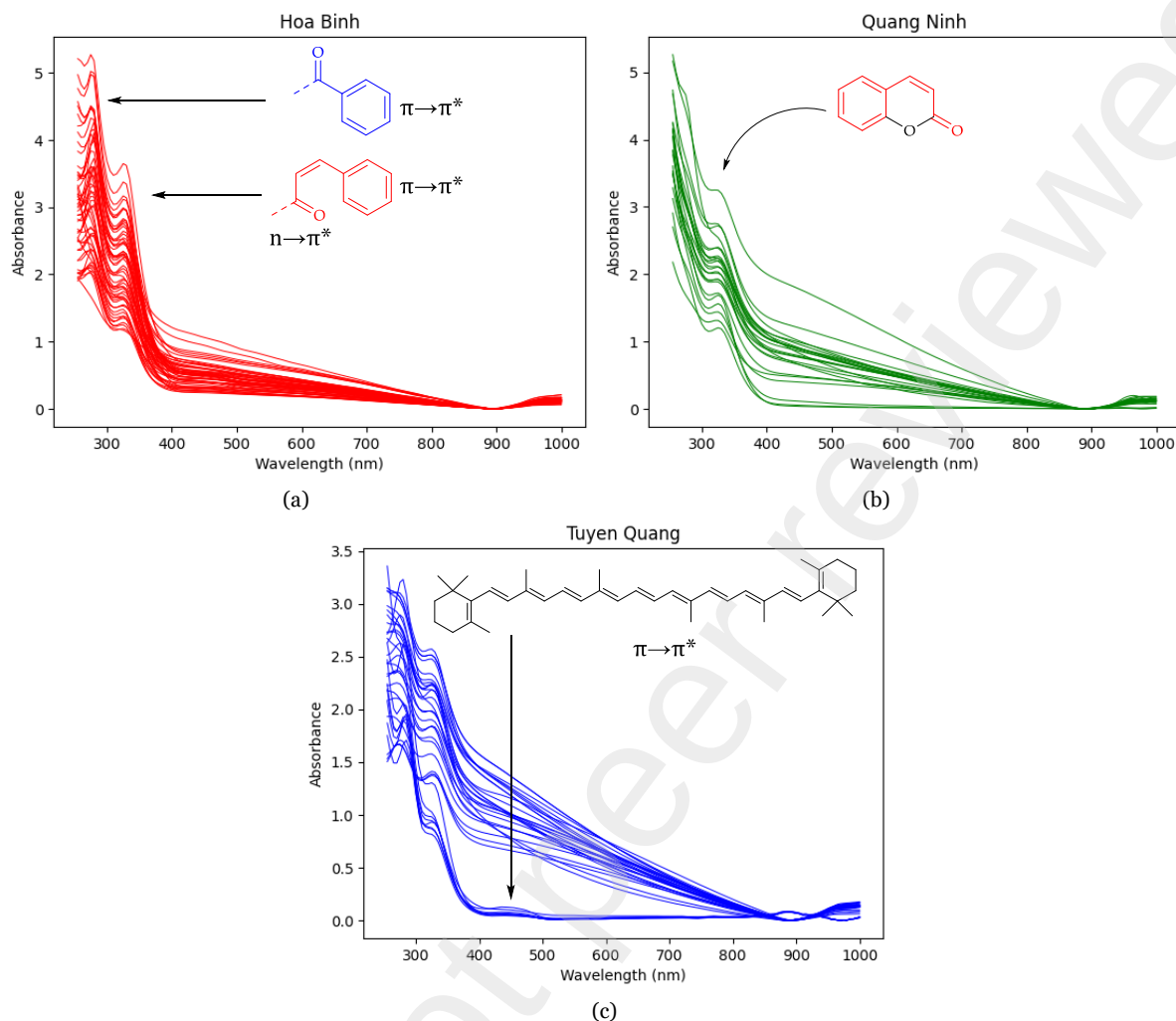


Fig. 4. Raw UV-Vis absorption spectra of orange juice samples from three different provinces of Vietnam: (a) Hoa Binh, (b) Quang Ninh, and (c) Tuyen Quang.

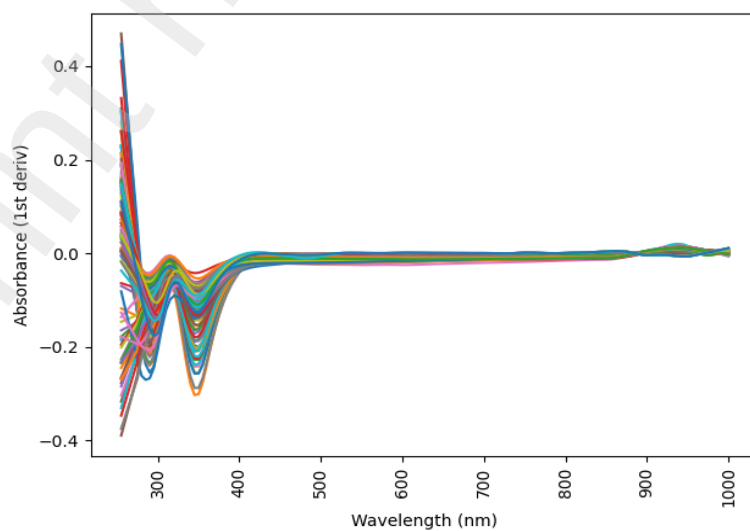


Fig. 5. Absorption spectra of the collected dataset after preprocessing with SNV + SG + FD.

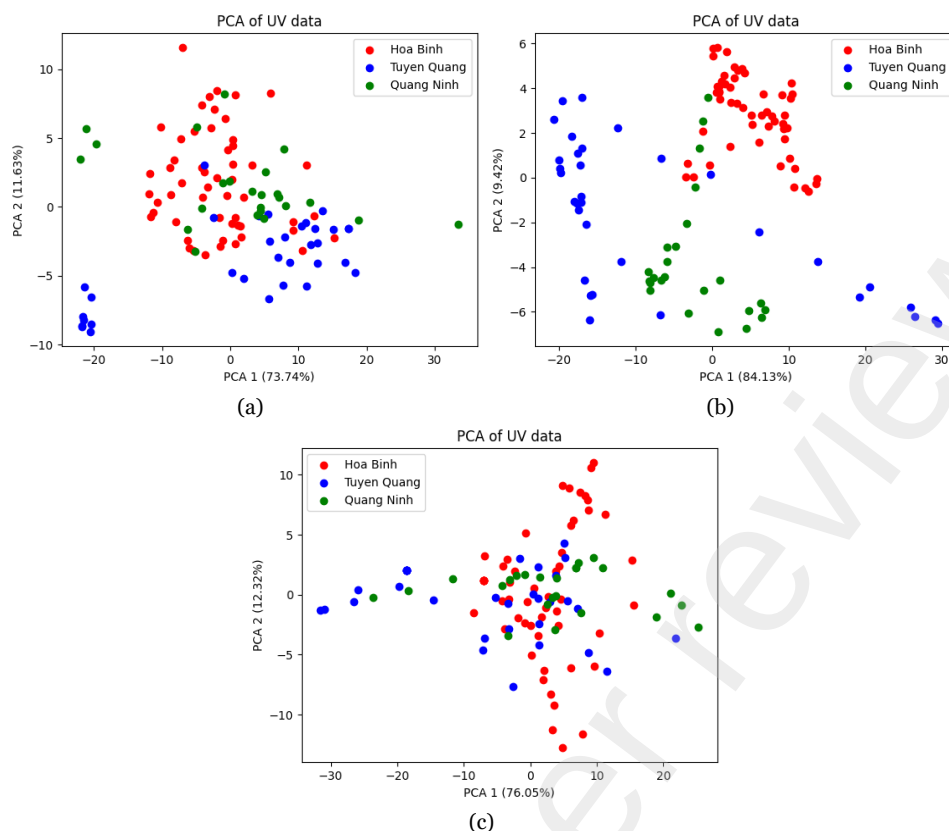


Fig. 6. PCA score plots of the UV-Vis spectral dataset for orange samples on the first two principal components, based on different preprocessing strategies: (a) raw data, (b) SNV + SG, and (c) SNV + SG + FD.

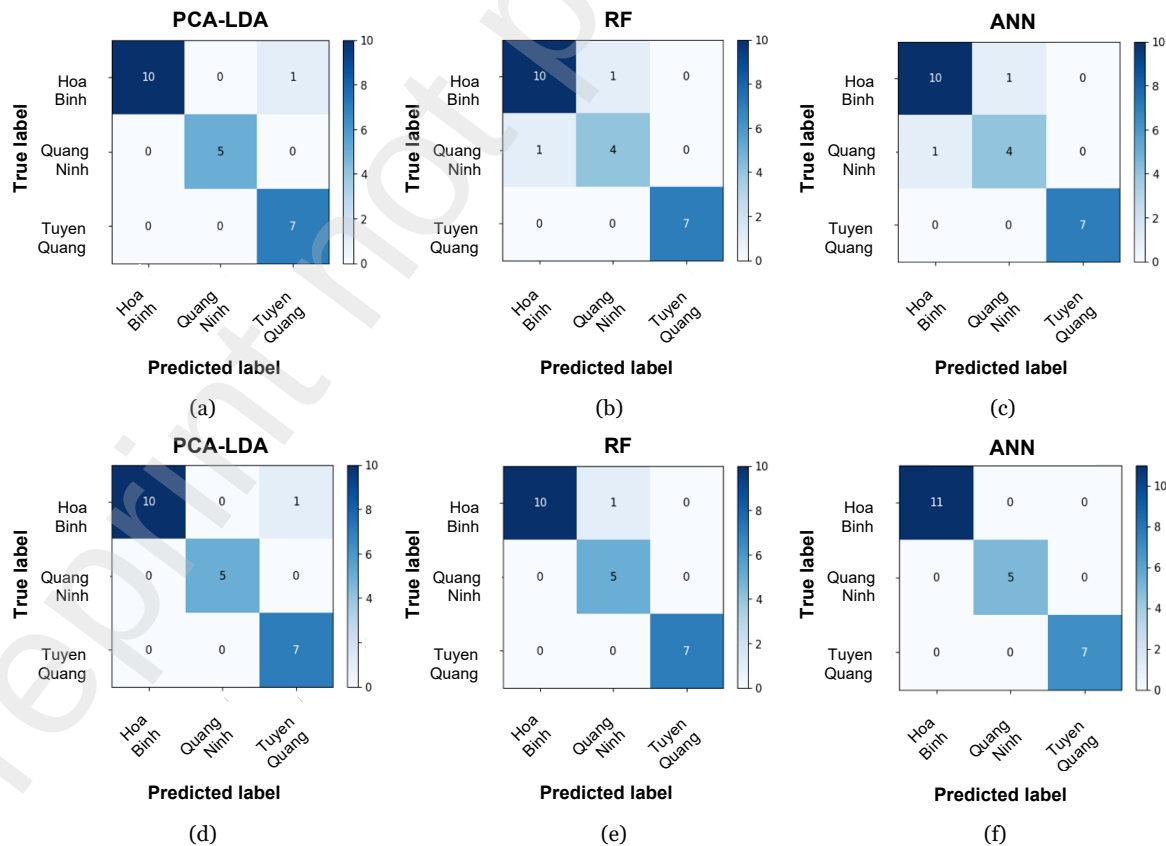


Fig. 7. Confusion matrices of different models using smoothed data with SNV and SG (a-c), and with an additional first derivative (SNV + SG + FD) (d-f): (a, d) PCA-LDA; (b, e) RF; (c, f) ANN.

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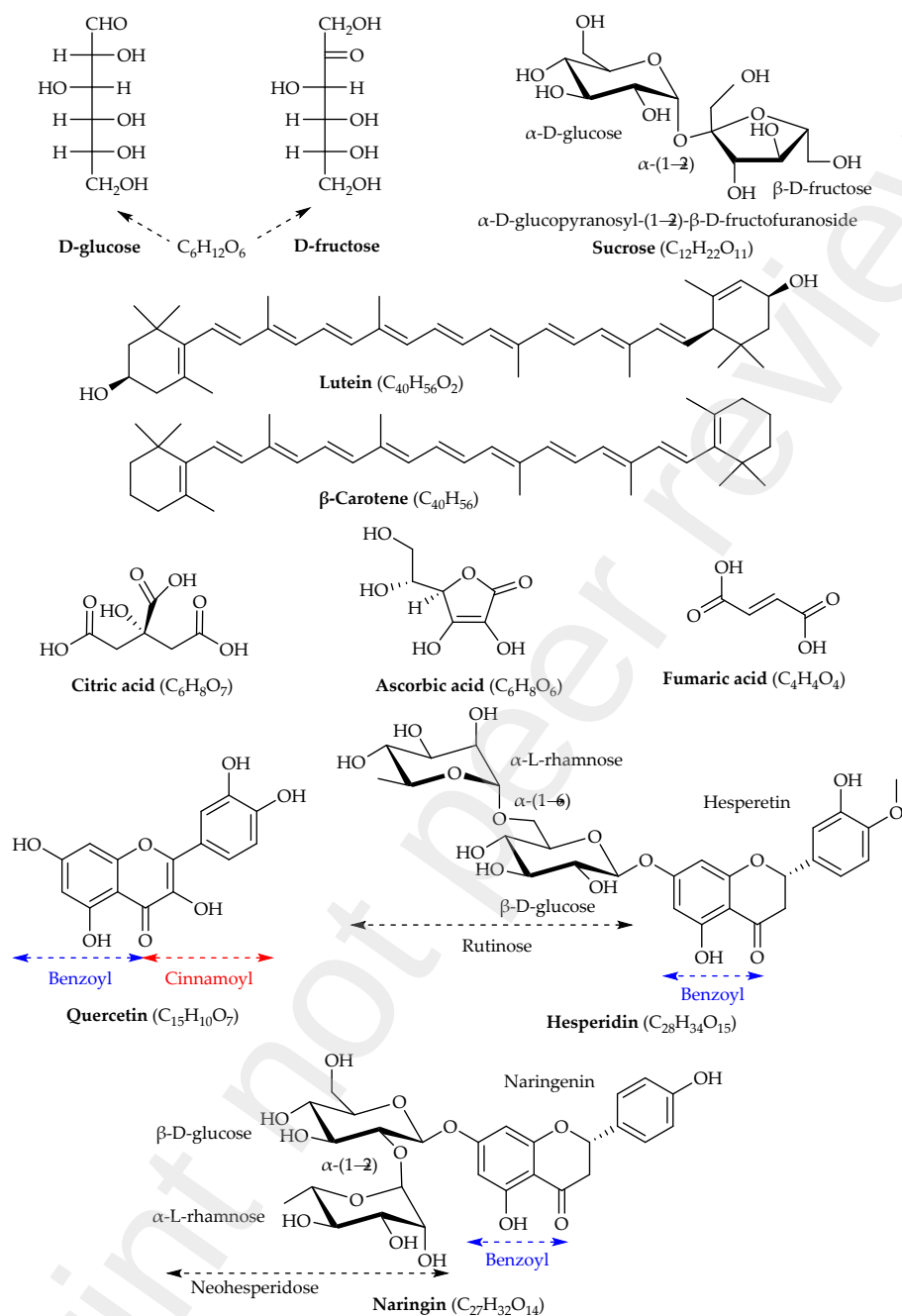


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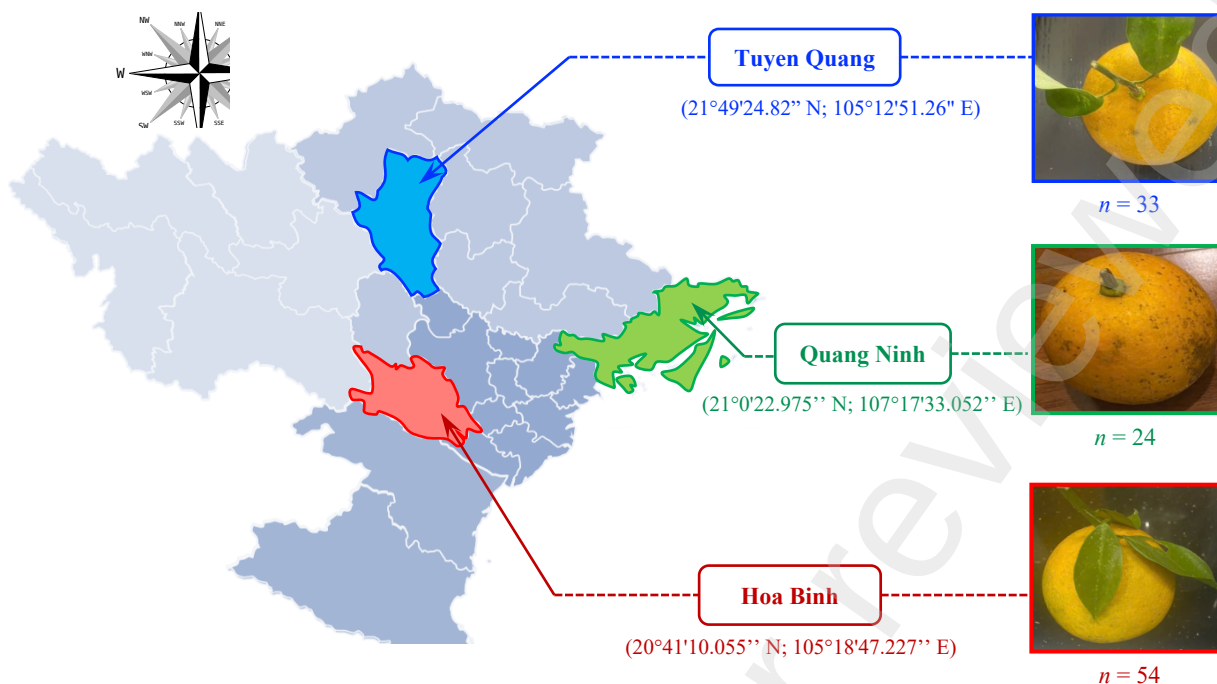


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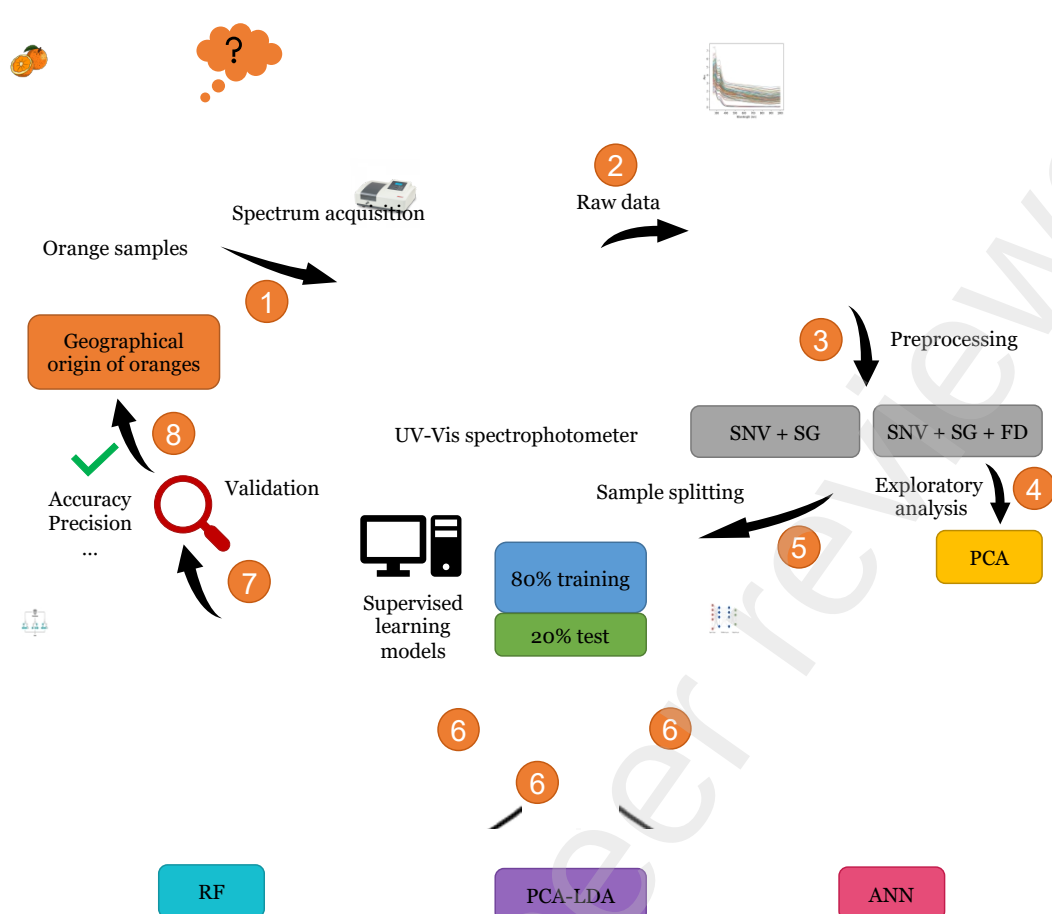


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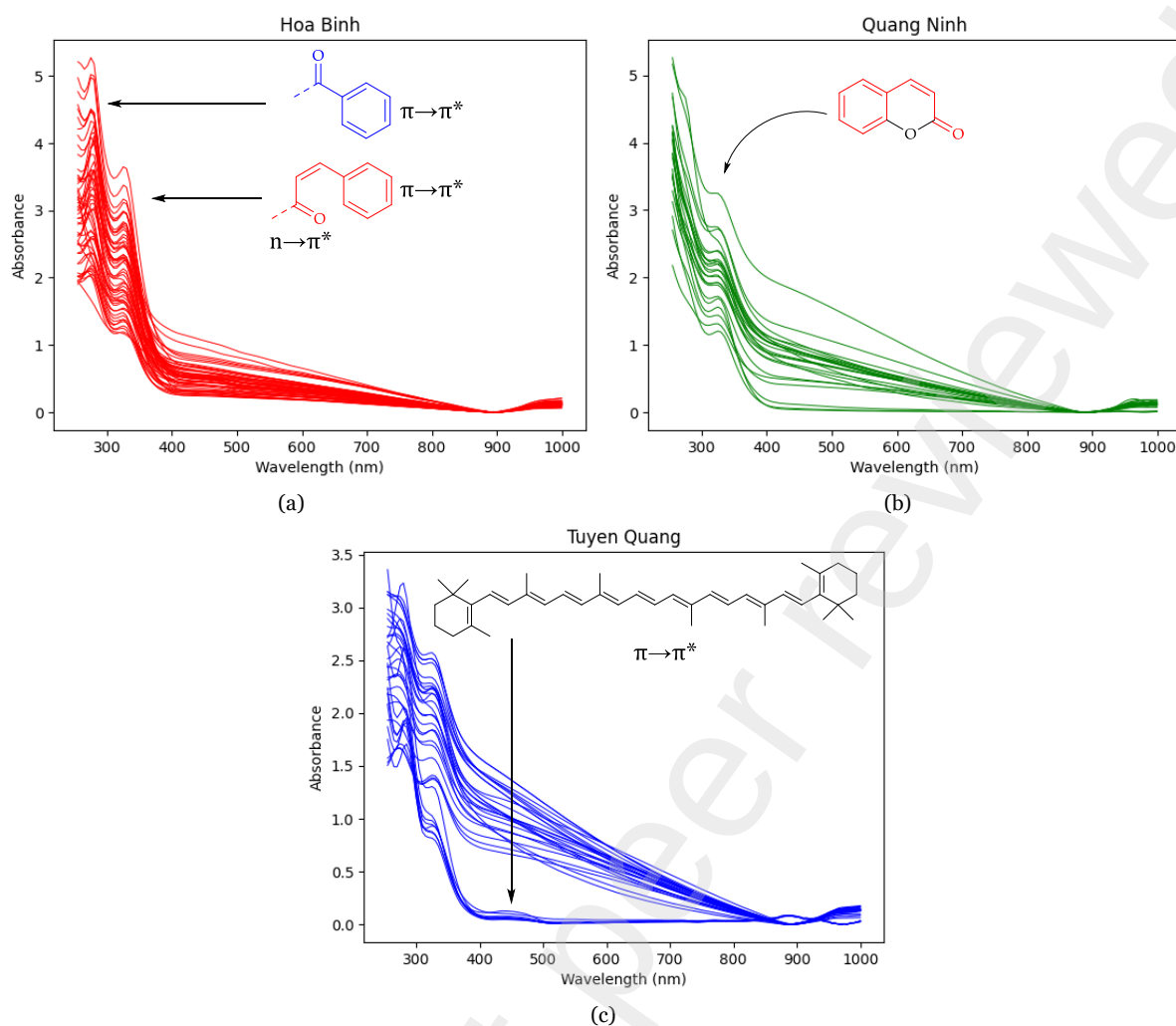


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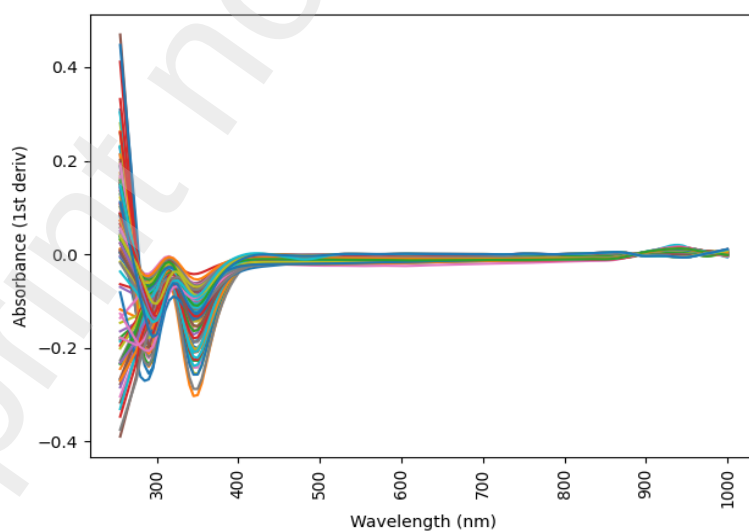


Fig. 5. Absorption spectra of the collected dataset after preprocessing with SNV + SG + FD.

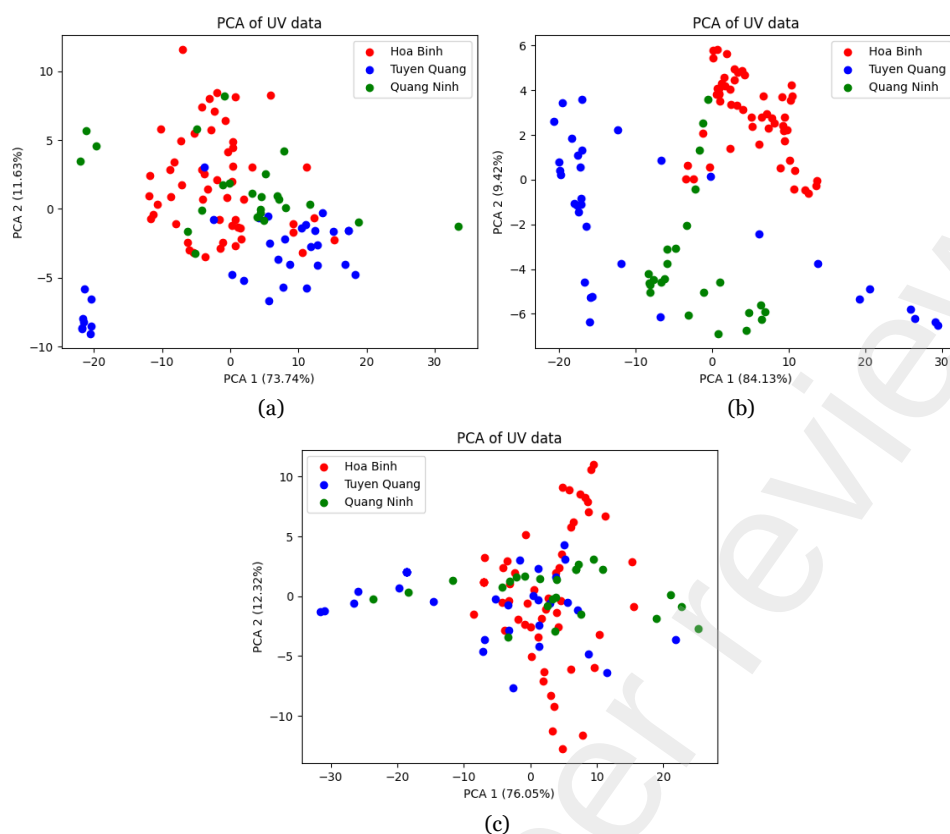


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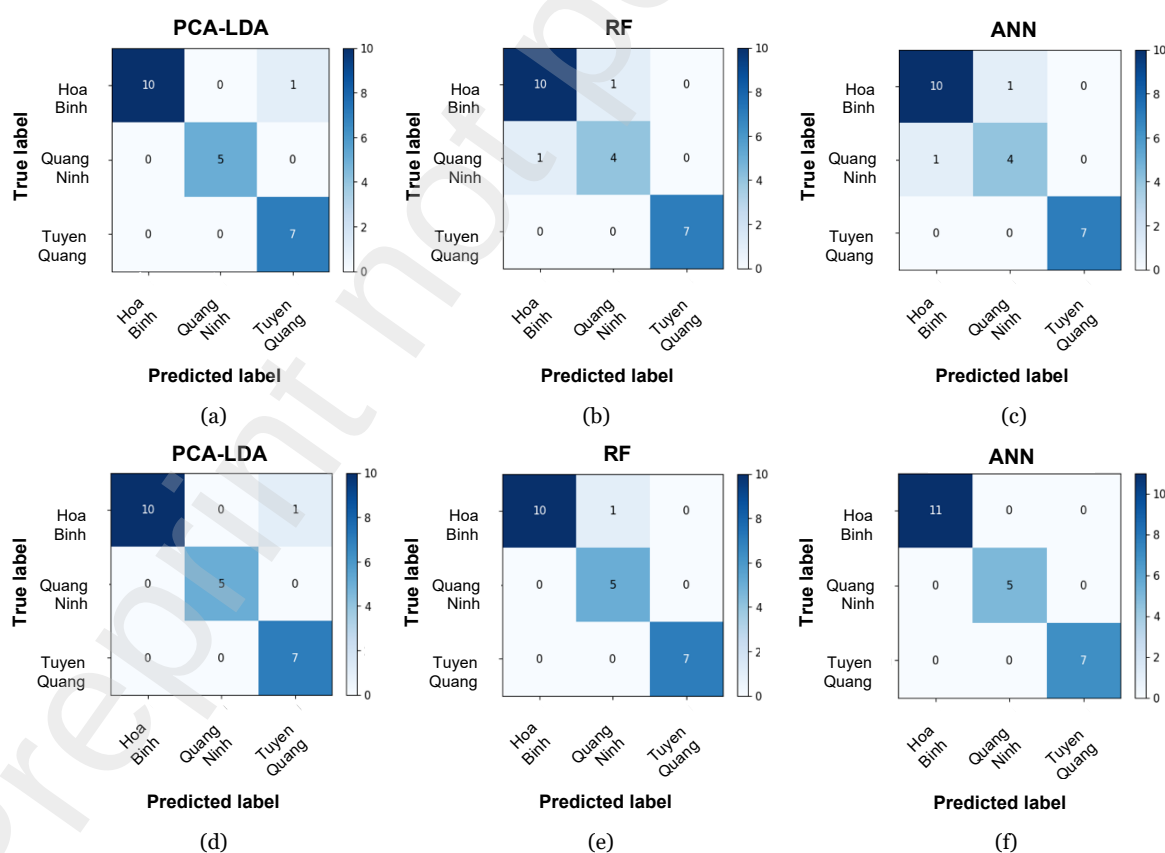


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List of Tables

Table 1. Classification performance parameters (%) of PCA-LDA, RF, and ANN models

Wavelength range (nm)	Pretreatment	Performance evaluation (%)	Classification models		
			PCA-LDA	RF	ANN
250-400	SNV + SG (order = 0)	A_{test}	100	86.9	86.9
		A_{training}	98.8	100	95.4
250-1000	SNV + SG (order = 0)	A_{test}	100	91.3	95.6
		A_{training}	98.8	100	97.7
		E	4.40	8.70	8.70
		P	96.5	90.9	90.9
		R	97.7	90.9	90.9
		S	98.8	90.9	95.6
		$F1$	96.5	90.9	90.9
	SNV + SG + FD (order = 1)	A_{test}	95.6	95.6	100
		A_{training}	96.5	100	100
		E	4.40	4.40	0
		P	96.5	94.4	100
		R	97.7	97.7	100
		S	98.8	98.8	100
		$F1$	96.5	95.6	100